Clean Data, Honest Numbers

Most reporting errors do not announce themselves. The query runs, the report arrives and the number looks reasonable, but it is wrong. This handout explains where these errors originate, how they reach reports, and how data professionals prevent them.

Where errors enter

Each step between data entry and a finished report can alter the data. Cleansing is the only step designed to detect and correct those changes.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Entry	**Export**	**Load**	**Cleanse**	**Report**
Typos, blanks, codes and local conventions	Exports keep text only; data types are lost	Software infers types without asking	Documented rules detect and correct errors	Filters, joins and counts assume clean input

Small errors compound

EACH FIELD

97%

of values are filled in

EVERY FIELD AT ONCE

45%

of records have no blanks

In the course data, each field is blank only about 3% of the time, yet just 45% of the 100,000 records are complete in every field. When each field carries its own small, independent chance of a gap, the chance that an entire record is complete is roughly the product of those chances:

```
0.97 × 0.97 × … × 0.97 (26 fields) ≈ 0.45
```

At 97% per field, five fields leave 86% of records complete; twenty leave 54%. The more fields a report depends on, the more cleansing matters.

Ten common problems in source data

Source systems are built for daily operations, not analysis. Each system, vendor and era has its own conventions, and merged data inherits them all. Expect several of these in any file; examples come from the course's fictional records.

01 Whitespace that looks blank

`"···"` `"·→·"` · space → tab

CAUSE Legacy systems pad fields to a fixed width, and exports preserve the padding.

IMPACT Padded values are not counted as missing, and each variation becomes its own category: 19 distinct genders in the course data instead of 3.

02 Placeholder codes for "unknown"

`N/A` `U` `UNK` `99:99`

CAUSE Many systems cannot store a true blank, so a placeholder, called a sentinel value, is entered instead.

IMPACT Placeholders are counted, sorted and averaged as real values. One exception: NONE in the housing column is a valid answer, not a missing one.

03 One fact, many formats

`06/18/1978` `14-MAR-2017` `20210117`

CAUSE Different screens, vendors and decades recorded dates and money in their own formats; merged data keeps them all.

IMPACT Text sorts character by character, not by date or amount, so "earliest" and "total" are calculated incorrectly.

04 IDs converted without a check

`02381457` → `2381457`

CAUSE Tools convert the ID to a number on their own, without checking that each value is a valid 8-digit ID.

IMPACT Superman's 02381457 becomes 2381457. Our tables store IDs that way on purpose, but a lookup or join using the text form now finds nothing.

05 One name, many spellings

`Kent` `kent` `kEnT`

CAUSE Free-text entry, inconsistent capitalization and different entry rules across screens.

IMPACT Exact matches miss variants. A search for KENT found none of Clark Kent's three records, and MCDONALD appears in 53 spellings.

Ten common problems, continued

06 Records are not people

`1 SID` → `3 offender numbers`

CAUSE A person keeps one SID for life but receives a new offender number with each incarceration.

IMPACT Counting rows as people inflates totals. In the course data, 8,499 people have more than one record.

07 Two-digit years and impossible dates

`2/19/64` → `2064-02-19`

`1213-03-22`

CAUSE Two-digit years saved keystrokes; software then picks the century by a fixed cutoff that ignores context.

IMPACT Ages come out negative or over 100. The course data needed 1,633 birth-date centuries corrected, and Wonder Woman is on file as born in 1213.

08 Two conventions in one column

`Y` `N` `T` `F` `y` `t` `U`

CAUSE Records from a legacy system (Y/N) and a newer system (T/F) were merged into one column.

IMPACT A filter for "Y" misses every T, y and t, counting 5,168 of the 6,878 records that should match.

09 Several values in one cell

`FLIGHT;SUPER STRENGTH`

CAUSE The entry screen offered a single field, so users typed lists into it.

IMPACT You cannot reliably count how many people on file can fly until each power has its own row.

10 Free text

`"no good reason given"`

CAUSE Officer notes capture what coded fields cannot, in each writer's own words and spelling.

IMPACT Keyword searches match the wrong notes and miss the right ones. A first search for "good" scored 0.12 on a scale where 1.0 is perfect.

Loud problems vs. quiet problems

Loud problems halt a process with an error, so they get fixed. **Quiet** problems let the process finish and produce a wrong answer. Nearly every problem on these two pages is quiet, so professionals check the data rather than wait for an error.

How bad data becomes a wrong number

Six mechanisms turn the problems on pages 2 and 3 into incorrect results. None of them produces an error message.

Silent conversion

Software converts data types as it loads, without asking. It turns ID text into numbers and yes/no text into true/false by rules that don't know your data.

IN THE COURSE

`bool("N")` is True, so a naive conversion marked all 96,744 filled-in records as protective custody.

Filters that miss variants

A filter matches exactly what you typed and nothing else. Every unmatched spelling, capitalization or code silently drops out of the result.

IN THE COURSE

= `"Y"` found 5,168 of 6,878 records;
= `"KENT"` found 0 of 3.

Blanks handled three ways

A blank can be skipped, counted as a value, or treated as zero, and each choice gives a different answer. An average that treats blanks as zero is pulled down.

IN THE COURSE

Missing custody levels: 1,536 as loaded, 2,991 once whitespace counts as blank.

Placeholders in the math

Codes such as 99:99 or 12/31/9999 look like real values, so they are sorted, averaged and charted as if they were.

IN THE COURSE

1,976 intake times are 99:99, the legacy system's code for "unknown."

Joins that drop or multiply rows

A join matches rows in two tables by a key. If keys don't match exactly, rows drop out. If a key repeats, every match repeats and totals multiply.

`1 person` `JOIN` `3 records` = `3 rows`

IN THE COURSE

A text key, 02381457, never matches a number key, 2381457.

Counting the wrong thing

Every percentage has a denominator. Records or people? Unknowns included or excluded? Change the denominator and the answer changes, with no error.

IN THE COURSE

100,000 records are not 100,000 people: 8,499 people have more than one record.

Rule of thumb: when a number surprises you, suspect the data before the world. When it doesn't, verify it anyway.

Seven dimensions of data quality

DAMA, the data management professional association, defines six dimensions of data quality. This course defines seven, each tested rather than judged by impression.

> **Why seven: validity split in two, because our records are decades old**
> Our data spans generations of systems and standards, so many values are right but in a retired shape: a two-digit year, a code missing its zero, Y/N where the newer system writes T/F. Lumped together, they bury the values that are actually wrong. The fixes differ too: a **conformity** problem gets a standardization rule; a **validity** failure goes to the data owner.

DIMENSION	HOW TO TEST IT	IN THE COURSE DATA
Completeness *Is everything there?*	Count true blanks after stripping spaces.	About 3% of each field is blank; 45% of records have none.
Conformity *Is it in the right shape?*	Match today's format: pattern, length, case.	`2/19/64` is a retired date format; `9B` lost its zero (`09B`).
Validity *Is it an allowed value?*	Check allowed codes, ranges and rules.	`V4` isn't a code, `IN TRANSIT` isn't a unit, `25:10` isn't a time.
Consistency *Does it agree everywhere?*	Compare across columns, records and systems.	8 people's birth dates conflict; 228 offense/violence codes clash.
Accuracy *Does it match reality?*	Check a trusted source or reasonable limits.	`2071-06-06`: right shape, real date, hasn't happened.
Uniqueness *Is each thing recorded once?*	Count repeated keys.	Offender numbers never repeat; SIDs repeat by design.
Timeliness *Is it current enough?*	Compare the as-of date with the need.	Every report should state its data's as-of date.

Cleansing vs. transformation

Cleansing

Makes values correct and consistent, never changing their meaning: trim, standardize, map codes, flag failures.

`"·kEnT·"` → `KENT`

Transformation

Reshapes data for a purpose: convert types, derive fields such as age at sentencing, split cells, join, summarize.

`FLIGHT;SUPER STRENGTH` → `2 rows`

Coded values and sensitive data

Units, race, religion, gangs and offense codes each have a few real answers and many ways to write them. The course data carries damage in seven domains:

DOMAIN	AS WRITTEN	STANDARDIZED
Names	`GARCÍA` `garcia` `MUÃ‘OZ`	`GARCIA` `MUNOZ`
Locations	`GOTHAM CITY` `Gothem City` `GOC`	`Gotham City · GOC`
Race and ethnicity	`W` `Caucasian` `W - WHITE`	`W · White`
Religion	`RC` `R.C.` `CATHLIC`	`Catholic`
Gang (STG) affiliation	`SUSP: H.Y.D.R.A.` `HYD (S)`	`HYDRA · Suspected`
Offense and violence	`13a` `9B` `V-2` `V4`	`13A` `09B` `V2` V4: no such code
Free text	`3rd shift` `wasnâ€™t`	`3rd shift` `wasn't`

A crosswalk, not a guess

An agreed reference table in Git maps each spelling to a standard value. `"W"` alone finds 14,069 records; the crosswalk, 28,765.

Unknown values go on a list

Values no crosswalk knows are listed by record ID, never guessed: 1,966, such as V4, IN TRANSIT (not a unit) and ASSAULT (13A or 13B?).

Same word, different meaning

In the gang field N/A means no known affiliation, a real answer; in religion, not recorded. A default load erased about 25,700.

Two facts in one field

Gang affiliation holds a group and a status. A bare name is Unverified, never Confirmed: `"HYDRA"` finds 113 of 1,608 HYDRA records.

Rules across columns

A crime against a person should carry a violent code. 228 records disagree; the data can't say which is wrong, so they're flagged.

Rules across records

Race shouldn't change between a person's records; 113 people have two codes. Religion may change, because people change faith.

Sensitive attributes

Race and religion are collected for required reporting and for accommodations. Standardize what was recorded, restrict who can see it, and never infer either from a name, a photo or anything else. Race maps to the agency's letter code and to the 2024 federal (OMB) categories. In the course data both are random and the gangs are fictional.

Eight practices of trustworthy cleansing

Tools change; these practices do not. Each one is demonstrated in the course.

1 **Load everything as text first** — SESSION 2

Read every column as text, then assign data types deliberately, so nothing is converted without your knowledge.

2 **Profile before you correct** — SESSIONS 1 AND 3

Count values and patterns to establish what the data actually contains before writing any rules.

3 **Preserve the original** — SESSION 2

Add cleaned columns alongside the originals and retain the raw file, so every correction can be reviewed and reversed.

4 **Write rules, not guesses** — SESSION 3

Use explicit formats and code lists. A tool's guess can change between versions; Wonder Woman's birth date parses differently in pandas 2 and 3.

5 **Let the data owner define meaning** — SESSION 2

Whether a blank, NONE or U means no, unknown or not applicable is a business decision, documented in a data dictionary.

6 **Flag, don't delete** — SESSION 4

List failed values by record ID so they can be fixed at the source. Some prove correct: the Winter Soldier really was 107 at sentencing.

7 **Make it repeatable** — SESSION 5

The same input always produces the same result, and running it twice changes nothing. Engineers call this idempotent; it lets a pipeline run unattended.

8 **Reconcile before you publish** — SESSION 5

Before anyone relies on a result, reconcile it: rows in match rows out, IDs stay unique, totals match to the cent, and a second method agrees.

How data moves

A pipeline is the set of automated steps that carries data from the system that creates it to the table a report reads, the same way every run.

STEP 1
Arrive
Extract, file or live feed

STEP 2
Land
Raw copy kept as received

STEP 3
Check
Counts and formats checked

STEP 4
Cleanse
Written rules applied

STEP 5
Publish
Trusted table saved

MONITOR EVERY RUN Each run is logged. A failure, a late file or a sudden jump in row counts alerts a person instead of waiting for a report to look wrong.

Where data arrives

Database extract
A scheduled query copies tables out of a source system, often nightly.

WATCH FOR
Columns renamed when the source is upgraded.

File drop
A system drops a file into storage, such as a cloud bucket.

WATCH FOR
Late, missing, partial or duplicate files.

Live feed
Events stream in as they happen, one message at a time.

WATCH FOR
Events arriving late, twice, or out of order.

ETL vs. ELT: when the cleaning happens

ETL Extract > Transform > Load

Data is cleaned in flight and only the result is stored, so the original is often lost.

ELT Extract > Load > Transform

Raw data lands first and is cleaned inside the platform, so the original stays available.

Why ELT helps trust: the raw copy is kept, so any number can be traced to what arrived, a corrected rule can be rerun over old data, and an auditor can check the work. Tools like Informatica PowerCenter were built for ETL; Databricks is built for ELT.

From raw data to a trusted table

Three layers: the medallion pattern

Most modern platforms, Databricks included, organize a pipeline in three layers.

LAYER 1
Bronze
Raw, as received, loaded as text. Nothing fixed, nothing lost.

LAYER 2
Silver
Cleansed and typed: one row per record, ID enforced.

LAYER 3
Gold
Ready for decisions: counts, rates and dashboards.

In the course: the raw CSV is bronze. Sessions 2 through 4 build silver, and session 5 saves it as a governed table. The Trust Test answers are gold.

Code in Git, data in governed storage

Code: Git
Notebooks and cleansing rules are versioned in Git, so every change records who made it, when and why, and can be rolled back.

Data: governed storage
Files and tables stay in governed storage with access controls, such as Unity Catalog. Real data never goes in Git, not even a sample.

In the course: the repository holds only code. The 100,000 mock records are generated from seed 2026 every time the setup cell runs.

Who does the work

- **Data engineer** builds and runs the pipelines that move, cleanse and check data.
- **Data analyst** turns clean data into answers, reports and dashboards.
- **Data steward** owns what the data means and the rules for its quality; usually a business expert, not IT.

Most cleansing work is conversation. Writing the code is the easier part; reaching agreement on what the data means is the real work.

Before you trust a number

Seven questions to ask of any report

- [ ] Where did the data come from, and as of when?
- [ ] What does a blank mean in each column I'm using: no, unknown, or not applicable?
- [ ] Am I counting records or people?
- [ ] How many rows went in, and how many came out?
- [ ] What didn't convert or match, and where is that list?
- [ ] Did my filters catch every spelling and code, such as Y and T, or Kent and KENT?
- [ ] Would a second method give the same number?

Words to know

Null The absence of a value. Not the same as spaces, zero or the text NONE.

Sentinel value A placeholder that stands for unknown, like 99:99 or N/A.

Data type What kind of value a column holds: text, whole number, decimal, date, or true/false.

Record ID (key) The value that identifies one record, like an offender number. Keys connect tables only when both sides store them the same way.

Join Matching the rows of two tables by a key.

Profiling Counting values and patterns to learn what a column really contains.

Data dictionary The agreed meaning of every column and code, owned by the business.

Pipeline Automated steps that move data from a source to a trusted table, the same way every run.

Landing zone Governed storage where raw data arrives before any cleaning, such as a cloud bucket or volume.

ETL / ELT Clean before loading, or load raw first and clean inside the platform.

Reconciliation Proving two results agree: row counts, unique IDs and totals.

Mock data only. Everything in this course is fictional. Never load real offender data into any tool that isn't approved for criminal justice information.